

Reversible Occurrence Nets

Hernán Melgratti

Daniel Dávalos

May 11, 2026

Contents

1	Nets	2
2	Occurrence nets	4
3	Reversible occurrence nets	6
3.1	Reversible Nets	6
3.2	Trace equivalence	7
4	Parabolic	8
4.1	Swapping	8
4.2	Parabolic lemma	8

Chapter 1

Nets

Definition 1 (Net). A *net* N is a tuple $N = (\alpha, \beta, \bullet_, \bullet\bullet)$, where α is a nonempty set of places, β is the set of transitions, and $\bullet_, \bullet\bullet : \beta \rightarrow \mathbb{N}^\alpha$ assign source and target to each transition. We consider only nets in which every transition both consumes and produces, i.e., $\bullet t \neq \emptyset$ and $t\bullet \neq \emptyset$ for all $t \in \beta$.

Definition 2 (Preset for places.). A *preset for places* is a function that take a net N , a place p and return the set $\{t \mid p \in \bullet t\}$, where the postset function is defined in Definition 1.

Definition 3 (Preset for transitions.). A *preset for transitions* is a function that take a net N , a transition t and return the multiset $\bullet t$.

Definition 4 (Postset for places.). A *postset for places* is a function that take a net N , a place p and return the set $\{t \mid p \in \bullet t\}$, where the preset function is defined in Definition 1.

Definition 5 (Postset for transitions.). A *postset for transitions* is a function that take a net N , a transition t and return the multiset $t\bullet$.

The notation for preset of places or transition are written as $\bullet_{\{\{N\}\}}t$, but in this file we'll use only the short notation $\bullet t$.

Definition 6 (Minimal). This definition say if the multiset doesn't have any transition before.

Definition 7 (Initial). A set s is *initial* in a net N if $s \subseteq \text{minimal}(N)$.

Definition 8 (Maximal). This definition say if the multiset doesn't have any transition after.

Definition 9 (Enabled transition). Given a net N and a multiset m , a vericator of enabled transition is a function $\text{is_enabled} : N \rightarrow \text{Multiset} \rightarrow \beta \rightarrow \text{Prop}$ that verifies if $m \leq \bullet t$. We denote $\text{is_enabled } N \ m \ t$ as $m[[t]]_{\{\{N\}\}}$.¹

Definition 10. Given a net N , a transition t and a prove that t is enabled at m , the function $\text{marking_after_firing } N \ m \ e$ returns the multiset applied at m on t , which is $m - \bullet t + t\bullet$.

Definition 11 (Firing). Given a net N , $\text{is_firing} : N \rightarrow \text{Multiset } \alpha \rightarrow \mathbf{h} \rightarrow \text{Multiset } \alpha \rightarrow \text{Prop}$ checks if a firing on a multiset m by t is equal to another multiset m' , i.e., if $m[[t]] = m'$. We denote this only by $m[[t]]m'$

¹In this documentation we avoid the subscript notation, and instead we write only $m[[t]]$.

Definition 12 (Firing sequence). `firing_sequence` $N \ m \ ls \ m'$ is the concatenation of sequences, where ls is the sequence of transitions, m is the initial marking (Definition 7) and m' the final marking of the sequence. We denote for $s = \varepsilon$ for a empty sequence, in Lean this is an empty list, we denote for $';$ ' the concatenation of lists. Therefore, a firing sequence is defined inductively as

- **Empty.** $m[[\varepsilon]]m$
- **Inductive step.** For a sequence $t; s$ where t is a transition and s is a transition sequence, $m[[t; s]]m'$ is a firing sequence if there is m'' such that $m[[t]]m''$ is firing and $m''[[s]]m'$ is a firing sequence.

Lemma 13 (Firing deterministic). *For a net N and a transition t enabled at m (where e is a proof that it is enabled), if $m[[t]]m'$ and $m[[t]]m''$ then $m' = m''$.*

Proof. Immediate by unfolding Definition 11. □

Lemma 14. *For every net N and multisets m, m' , $m[[\varepsilon]]m'$ implies $m = m'$.*

Proof. By structural induction on the firing sequence. □

Lemma 15. *Let t be a transition and s a sequence (note that $t; s$ is a sequence), m, m' two multisets and $m[[t; s]]m'$, then there exist a multiset m'' such that $m''[[s]]m$.*

Lemma 16. *Let t be a transition, s a sequence of transitions, m, m' two multisets, and $m[[t; s]]m'$. Then, there exist a multiset m'' such that $m[[t]]m''$.*

Lemma 17 (Concat firing sequence). *Let be $m[[s_1]]m'$ and $m''[[s_2]]m''$ two firing sequences. Then $m[[s_1; s_2]]m''$.*

Lemma 18 (Append split of firing sequence). *Let $m[[s_1; s_2]]m'$ be a firing sequences, then there is a multiset m'' such that $m[[s_1]]m''$ and $m''[[s_2]]m'$.*

Proof. By induction on the length of s_1 , and by structural induction on the firing sequence for s_2 . □

Lemma 19 (Firing sequence deterministic). *Let be $m[[s]]m'$ and $m[[s]]m''$ two firing sequences, then $m' = m''$.*

Definition 20. A *marked net* is a tuple $M = (N, m_0)$, where N is a net (Definition 1) and m_0 is the initial marking.

Definition 21 (Reachable). Given a marked net M and a multiset m , *reachable* $N \ m$ return all the multisets that can be executed by sequences of firing enabled. We denote a reachable for a marking m of a net N by $N \rightsquigarrow m$.

Theorem 22 (Square lemma). *Let N a net, $m[[t_1]]m_1$ and $m[[t_2]]m_2$ two firings, and $\bullet t_1 \cap \bullet t_2 = \emptyset$. Then, there is a multiset m' such that $m_1[[t_1]]m'$ and $m_2[[t_2]]m'$.*

Proof. It suffices to take $m' = m - \bullet t_1 + t_1 \bullet - \bullet t_2 + t_2 \bullet$. To prove that $m_1[[t_1]]$ note that we can have $m_1 = m - \bullet t_1 + t_1 \bullet$ after firing, and $\bullet t_2 \leq m - \bullet t_1$. Analogously with the enabled $m_2[[t_2]]$. □

Chapter 2

Occurrence nets

Definition 23 (Immediate dependency). Let N be a net. The *immediate dependency relation* $<_N \subseteq (\alpha \cup \beta) \times (\alpha \cup \beta)$ is defined by the set

$$\{(x, y) \mid (y \in \beta \wedge x \in \bullet y) \vee (x \in \beta \wedge y \in x \bullet)\} \quad (2.1)$$

Definition 24 (Causal relation). We denote the transitive closure of $<_N$ as the *causal relation* $\prec_N$.

Definition 25. A relation $\preccurlyeq_N$ is *causal preorder* if it is the reflexive-transitive closure of $<_N$ (from Definition 23).

Definition 26. Let N be a net and $a \in \alpha$ a place. If $a \in \text{minimal}(a)$ and $x \neq a$ then $\neg(x \preccurlyeq a)$.

Definition 27. Two different transtions t, t' are in *immediate conflict* if $\bullet t \cap \bullet t' \neq \emptyset$. Let's denote this relation as $t \#_0 t'$.

Definition 28 (Conflict). Two elements $x, y \in \alpha \cup \beta$ are in *conflict* if there exist two transtions t, t' that $t \preccurlyeq x$, $t' \preccurlyeq y$ and $t \#_0 t'$. We denote a conflict relation of x, y as $x \# y$.

Definition 29 (Backward conflict). A place a has a *backward conflict* if it is produced by two distinct transitions, i.e., $\exists t_1, t_2, (t_1 \neq t_2) \wedge (a \in t_1 \bullet) \wedge (a \in t_2 \bullet)$.

Definition 30 (Acyclicity). A net is *acyclic* if its causal relation is irreflexive, i.e., there is no $x \in \alpha \cup \beta$ such that $x \prec_N x$.

Definition 31 (Occurrence net). A marked net (N, m_0) is an *occurrence net* if it satisfies the following statements:

1. N is acyclic;
2. there is no self-conflict, i.e., $\neg(t \# t)$ for all transition t ;
3. there is no backward conflict;
4. N is 1-safe, i.e., marking is a set;
5. the initial marking is the set of initial places, i.e., m_0 is minimal (Definition 6).

Definition 32 (Marked occurrence net). A marked net (N, m_0) is called *marked occurrence* if N is occurrence, m_0 has no duplicates and each place $p \in m_0$ implies $p \in \text{minimal } N$.

Definition 33 (Concurrency). Two different elements $x, y \in \alpha \cup \beta$ are *concurrent* in N if $\neg(x \prec y)$, $\neg(y \prec x)$ and $\neg(x \sharp y)$. We denote the concurrency between these elements as $x \text{ co } y$.

Definition 34 (Concurrent multiset). A multiset m is concurrent in N if it no has duplicate and

$$\forall x, y \in \alpha, (x \in m) \Rightarrow (y \in m) \Rightarrow x \neq y \Rightarrow x \text{ co } y.$$

We'll denote the concurrency of a multiset m by **Concurrent** m .

Lemma 35. *Let be N an occurrence net, **Concurrent** N m and $m' \leq m$, then **Concurrent** N m' .*

Lemma 36. *Let be N an occurrence net and **Concurrent** N $(m + t\bullet)$, then **Concurrent** N $(m + \bullet t)$.*

Lemma 37. *Let N be an occurrence net, t an enabled transition at m and a, b two different places. If **Concurrent** N m , $m[[t\rangle m'$ is a firing, and $a, b \in m'$, then $a \text{ co } b$.*

Lemma 38. *Let N be an occurrence net and t a transition enabled at m . If **Concurrent** N m and $m[[t\rangle m'$ is a firing, then **Concurrent** N m' .*

Lemma 39. *Let N be an occurrence net and s a transition sequence. If N is concurrent at m and $m[[s\rangle m'$ is a firing sequence, then **Concurrent** N m' .*

Proof. The proof procedes by applying structural induction over the length of the sequence s . For the base case apply the Lemma 14 and is obtained that $m' = m$, so it is concurrent at m' by hypothesis. For the step case, we can apply the inductive hypothesis over Lemma 38. $\square$

Lemma 40. *Every marked occurrence net (N, m_0) is marking concurrent at m_0 .*

Theorem 41. *Every marked occurrence net (N, m_0) is safe.*

Proof. A net is safe if for each multiset m and sequence s , $m_0[[s\rangle m$ implies that m has no duplicate tokens. So, let m be a multiset and s a transition sequence.

We've proved (Lemma 40) that m_0 is concurrent in N , so it is enough to apply the Lemma 39 to the occurrence condition in the hypothesis s , firing sequence $m_0[[s\rangle m$ and use the Lemma 40. $\square$

Chapter 3

Reversible occurrence nets

3.1 Reversible Nets

Definition 42 (Transition Type). Lets define the type $\text{Transition} ::= \text{fwd} \mid \text{bwd}$, where their elements are the forward direction for fwd and backward direction for bwd . Given a type for transition β , we define the set of reversible transitions as pairs $t_r ::= \langle t, \text{fwd} \rangle \mid \langle t, \text{bwd} \rangle$ with $t \in \beta$.

Definition 43. For $b \in \beta$, we'll say that a *transition* $t = (b, \text{fwd})$ is *forward* and let's denote it as $\triangleright t$.

Definition 44. For $b \in \beta$, we'll say that a *transition* $t = (b, \text{bwd})$ is *backward* and let's denote it as $\triangleleft t$.

Definition 45. Let s a sequence of transitions with types $\beta \times \text{Transition}$. We denote $\triangleright \triangleright s$ if forall $t \in s$, $\triangleright t$; or if $s = \varepsilon$.

Definition 46. Let s a sequence of transitions with types $\beta \times \text{Transition}$. We denote $\triangleleft \triangleleft s$ if forall $t \in s$, $\triangleleft t$; or if $s = \varepsilon$.

Definition 47.

Definition 48 (Reversing occurrence nets). A *reversible occurrence net* is a net $N = (\alpha, \beta \uplus \{\neg t \mid t \in \beta\}, \bullet, _, _ \bullet)$ defined by the following conditions:

1. $(\alpha, \beta, (\bullet _)_{|\beta}, (_ \bullet)_{|\beta})$ is an occurrence net, where $f_{|\beta}$ denotes the restriction of the function f to the domain β .
2. For every $t \in \beta$, $\bullet t = (\neg t) \bullet$ and $t \bullet = \bullet (\neg t)$.

Definition 49. For a reversible net R , if $m[[t]]$ and $m[[t]m']$ then there is a transition t' , which $t' \longleftrightarrow t$ and $m'[[t']]$.

Lemma 50 (Cancellation). *If $t \longleftrightarrow t'$, $m t \rangle m'$ and $m'[[t']]$ then $m' t' \rangle m$.*

Proof.

□

Definition 51 (Marked reversible occurrence net). A tuple $MO = (R, m_0)$ is *marked reversible occurrence* if R is a reversible net and $(\text{fwd_subnet } R, m_0)$ is marked occurrence, where fwd_subnet returns the forward subnet of a reversing net.

Lemma 52 (Loop). *Let $t, t' \in \beta \times \text{Transition}$ two reversing transitions, i.e., $t \longleftarrow t'$. Then,*

$$(\exists e, m[[e)m']) \iff (\exists e, m'[[e)m])$$

Theorem 53 (Backtracking). *If $(\exists s, m[[s)m'])$ then $(\exists s, m'[[s)m])$.*

Proof. By induction over the length of a sequence s . By applying the Lemma 52 to the step case $t; s$ on s , we can obtain a witness $s; \leftarrow t$ and apply the respective concatenation (Lemma 17) and we have the firing sequence as expected. $\square$

3.2 Trace equivalence

Definition 54 (Transition equivalence). The statement of the lemma requires a formal notion of *equivalent firing sequences*, which is the smallest congruence¹ $\asymp$ on transition sequences closed under the following rules:

- (i) $t; t' \asymp t'; t$ if $t \text{ co } t'$,
- (ii) $t; \leftarrow t \asymp \varepsilon$, and
- (iii) $\leftarrow t; t \asymp \varepsilon$.

Definition 55. $s \asymp s$.

Definition 56. $s_1 \asymp s_2 \implies s_2 \asymp s_1$.

Definition 57. $s_1 \asymp s_2 \wedge s_2 \asymp s_3 \implies s_1 \asymp s_3$.

Theorem 58. $s_1 \asymp s_2 \wedge s_3 \asymp s_4 \implies s_1; s_3 \asymp s_2; s_4$.

Proof. By cases over the structural definition 54. $\square$

¹A relation R over sequences is a **congruence** if it is an equivalence relation closed by concatenation, i.e., $s_1 R s_2$ and $s_3 R s_4$ implies $s_1; s_2 R s_3; s_4$ for all s_1, s_2, s_3 and s_4 .

Chapter 4

Parabolic

4.1 Swapping

Definition 59. Let $t \in \beta$ a transition and s a sequence of transitions of β . We define

$$t \notin^r := \forall t' \in s, \neg(t \longleftrightarrow t').$$

Lemma 60. Let $M = (N, m_0)$ a marked reversible occurrence net, m a marking, t and t' two transitions with $t \bullet \cap \bullet t' = \emptyset$ and $\neg(t \longleftrightarrow t')$. If $M \vdash m$ and $m[t; t'] \gg m'$, then $m[t'; t] \gg m'$.

Lemma 61. Let $M = (N, m_0)$ a marked reversible occurrence net, m a marking, t and t' two transitions with $\triangleright t$ (is forward transition) and $\triangleleft t'$ (is backward transition). If $M \vdash m_0$, $\neg(t \longleftrightarrow t')$ and $m[t; t'] \gg m'$ is a firing sequence, then we can have the swapped firing sequence between the transitions: $m[t'; t] \gg m'$.

Lemma 62 (Swapping). Let $M = (N, m_0)$ a marked reversible occurrence net, m a marking, s a sequence and t a transition with the condition $\triangleleft \triangleleft s$ and $\triangleright t$. If $M \vdash m_0$, $t \notin s$ and $m[t; s] \gg m'$ is a firing sequence, then we can have the swapped between the transition and all the backward sequence: $m[s; t] \gg m'$.

4.2 Parabolic lemma

Definition 63.

Theorem 64 (Parabolic lemma). Let R be a reversible occurrence net, m a reachable marking, and $m[[\omega] \gg m'$ a firing sequence. Then there exist two sequences of transitions, ω' and ω'' , such that ω' consists only of backward transitions, ω'' consists only of forward transitions, $\omega \preceq \omega'; \omega''$, and $m[[\omega'; \omega''] \gg m'$.

Proof. By applying induction on the firing sequence. The base case when $m' = m$ is immediate. The inductive step applies the property of $\preceq$ and the Lemma 63. $\square$